

Airframe Design Theory Exploration: Part II

Relevant Reading

- [Flight Vehicle Design Book](#)
 - Chapter 4: Wing Design (7 pages)
- [Computational Aeronautics](#)
 - Chapter 4: Finite Wing (33 pages)

Problem 1 Vocabulary

Explain the following terms; making sure to use sufficient detail, including any math or helpful figures. In some cases, these terms are simple one sentence definitions, in others, you should include several paragraphs to explain them fully.

Non-dimensional Numbers

- Reynolds Number
- Mach Number
- Coefficients
 - Lift Coefficient
 - Drag Coefficient
 - Moment Coefficient

Airframe Performance

- Lift Distribution
- Stall Speed

Problem 2 Theory

In this problem, you get to dive into the theory the vortex lattice method.

- i. What is potential flow theory? What is the governing equation? What are the assumptions?
- ii. What is a vortex filament? What is a horseshoe vortex?
- iii. How do you use horseshoe vortices to model a lifting surface? What type of solver do you need to use?
- iv. Explain why an external drag model is required to capture stall.

Problem 3 Exploration

Complete the following exploration.

3.a Lift Distributions and Wing Efficiency

- i. Create a plot comparing the lift distributions of a constant chord, tapered chord, and elliptic wing with no twist and with the same wing area. Also plot the ideal elliptic distribution for comparison.
- ii. Discuss the differences you see in your plot. Which planform design is the most efficient? Why?